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Data from Artificial Models of Mitochondrial DNA Disorders Are Not Always Applicable to Humans

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Mitochondrial DNA (mtDNA) mutations, a major cause of maternally inherited human diseases, are commonly characterized by the coexistence of mutant and wild-type mtDNA molecules within a cell (called "heteroplasmy" or "mutation load"). Usually, the higher the mutation load, the more severe the disease. Because of the high risk of recurrence in siblings and the absence of treatment options, couples at risk of transmitting mtDNA mutations often ask for prenatal diagnosis (PND) or preimplantation genetic diagnosis (PGD). These procedures are based on assessment of the mutation load (percentage of mutant mtDNA / total mtDNA) in either tissues (PND) or cells (blastomeres in a day 3 embryo). The predictive value of these procedures for postnatal outcome assumes that the mutation load is relatively stable among blastomeres at the 8-cell-stage embryo or in tissues throughout the entire embryofetal development.

By measuring mtDNA heteroplasmy levels in single blastomeres sampled from 8-cell-stage macaque embryos, Lee et al. reported a significant difference in heteroplasmy levels among cells from a given embryo, questioning the reliability of the PGD procedure in clinical practice (Lee et al., 2012). In their experiments, they removed half the cytoplasm of an oocyte derived from a female rhesus macaque and fused it with another halved oocyte containing the spindle-chromosomal complex recovered from a second female, thus generating artificially heteroplasmic embryos containing 50% of each mtDNA species. They observed a large variation of mtDNA mutation loads among blastomeres (mean of the standard deviations [SDs] of 15%, ranging from 3% to 28%). This feature increased dramatically during the first cleavages, and it was still present at the blastocyst stage. On the basis of these observations, Lee et al. concluded that heteroplasmy in one biopsied blastomere from cleaving embryos may not be predictive of total mutation load in the remaining blastomeres and the whole embryo. They therefore suggested that chorionic villus sampling could be more reliable than blastomere assessment for the prediction of embryonic mutation load.

Our experience of PGD for mtDNA mutations in humans does not support conclusions from these macaque experiments. Indeed, all studies on human heteroplasmic embryos hitherto conceived by in vitro fertilization have shown quite good stability of the mutation load among blastomeres, irrespective of the variant type: a mtDNA polymorphism (HV2 tract, Steffann et al., 2006) or mtDNA mutations (m.3243A>G, m.8993T>G, m.8344A>G, m.9185T>C, m.10197G>A, Steffann et al., 2006, Monnot et al., 2011, Treff et al., 2012, Vandewoestyne et al., 2012, Sallevelt et al., 2013). Figure S1C summarizes all published data and unreported personal observations regarding single-blastomere analysis on a total of 105 human embryos carrying various mtDNA mutations. At variance with the SD observed by Lee et al. (15%), the mean SD of the mutation loads was very low in our series (less than 2%), which can be ascribed to reproducibility of the technique (Steffann et al., 2006). In the less favorable cases, the SD was consistently below 15%, far less than the SD figure reported by Lee et al. (28%). For comparison between data from the macaque experiment and human clinical data, we plotted the rate of mutation in one randomly chosen blastomere (using a random number generator) against that in remaining cells of the same embryo (Figure S1). In rhesus macaque embryos, the test blastomere heteroplasmy was poorly predictive of the heteroplasmy observed in the remaining seven blastomeres (R^2 = 0.066, Figure S1A). Conversely, in human embryos, the test blastomere heteroplasmy was highly predictive of the mutation load in the remaining blastomeres (R^2 = 0.98, Figure S1B). The linear regression was more significant in human than in rhesus macaque embryos ($p < 0.0001$ and $p = 0.0013$, respectively).

Why results differ so markedly across species is an interesting question, especially given that mouse heteroplasmic embryos were reported to display very little variations in heteroplasmy levels among blastomeres (mean 2%, ranging from 0 to 6%, Dean et al., 2003). The difference may be due to the experimental procedure used to generate the heteroplasmic macaque model, which is likely to result in an mtDNA segregation pattern

different from that occurring in nonmanipulated embryos. Because such embryos are formed by the merging of two distinct cytoplasts, mitochondria from donors and acceptors may be clustered in different cell areas, leading to a nonrandom distribution of mtDNA populations between the two daughter cells. Such a hypothesis is supported by the results of another study, carried out in the mouse, using a microsurgery approach similar to that of Lee et al. in the monkey (Meirelles and Smith, 1998). The variation of heteroplasmy among blastomeres was 10-fold higher in reconstructed embryos than in zygotes derived from heteroplasmic lineage (Meirelles and Smith, 1998). In their cases, the supposed "rapid mitochondrial DNA segregation" was ascribed to a partitioning of mtDNA genotypes occurring in the process of fusing the two egg cells, a feature that probably does not occur in naturally heteroplasmic oocytes.

These observations emphasize the risk of mis-segregation of mtDNA molecules in "artificially generated embryos." Lee's artificial model of mtDNA disorders might thus be more relevant to preventing the transmission of mtDNA disorders by nuclear transfer than by embryonic selection using PGD. However, it has to be emphasized that small amounts of mutant mtDNA are usually cotransferred with the karyoplast during nuclear transfer from affected to donor oocyte (Tachibana et al., 2009, Craven et al., 2010). Even if rapid segregation of mutant mtDNA occurs, the risk induced by the resulting heteroplasmy can probably be regarded as negligible. Technical improvements have indeed decreased the carryover heteroplasmy to levels below 1%, far less than the threshold of mtDNA disease.

Taken together, our data, as well as those from other groups, advocate a homogeneous distribution of wild-type and mutant mtDNAs in individual human blastomeres at the 8-cell stage, regardless of the mutation. They support the reliability of PGD procedures based on mutation load assessment on two different blastomeres sampled at day 3 (Monnot et al., 2011, Sallevelt et al., 2013, Vandewoestyne et al., 2012). Moreover, stability of the mutation loads between the inner cell mass and the trophectoderm has been reported at the blastocyst stage (Treff et al., 2012), at least for the m.3243A>G mutation, supporting the feasibility of PGD on a day 5 trophectoderm biopsy. The four children born to date after a PGD procedure had buccal cell or cord blood mutation loads very similar to the blastomere mutation loads (m.8993T>G: 0% versus 0% in the embryos [Steffann et al., 2006], m.3243A>G: 15% versus 12% [Treff et al., 2012] and 5% versus 5% and 13% [two transferred embryos; Monnot et al., 2011], m.8344A>G: 63% versus 53% and 59% [two transferred embryos; this report]). Although more data are certainly needed in order to document the stability of the mutation loads throughout human in utero development, these observations support the reliability of standard PGD procedures on human embryos at risk of carrying mtDNA mutations.

SUPPLEMENTAL INFORMATION

Supplemental Information includes one figure and can be found with this article online at http://dx.doi.org/10.1016/j.celrep.2014.05.005.

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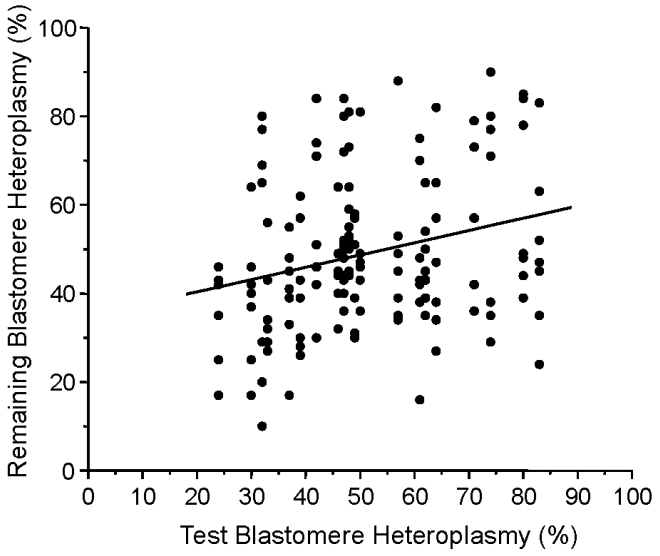
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Supplemental Information

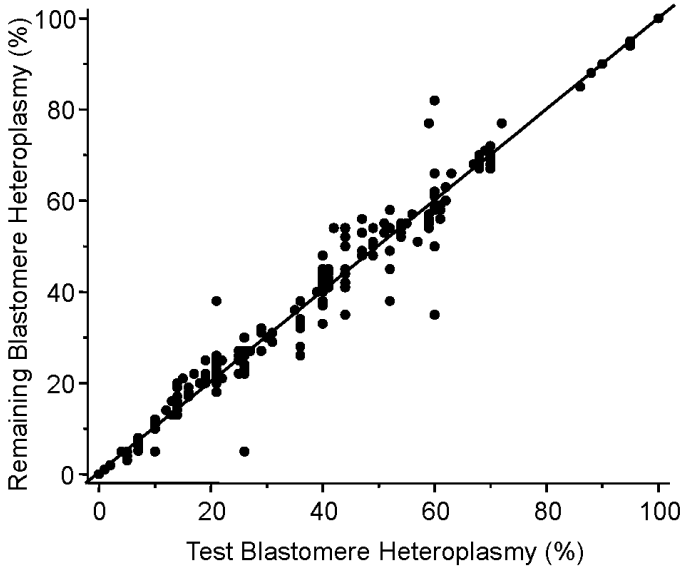
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(A) Rhesus Macaque (artificial heteroplasmy)



(B) Human (natural heteroplasmy)



(C) Mutation Load (%) in Blastomeres from Human Preimplantation Embryos

Embryos	Blas. 1	Blas. 2	Blas. 3	Blas. 4	Blas. 5	Blas. 6	Blas. 7	Blas. 8	Blas. 9	Blas. 10	Blas. 11	Blas. 12	Blas. 13	Whole Embryo	SD	References
MT-ATP6: m.8993T>G (*Gene*: Mutation)																
PIVc1e1	88	88												88	0	Sallevelt et al., 2013
PIVc1e3	0	0	0	0											0	Sallevelt et al., 2013
PIVc1e4	0	0	0	0	0	0	0	0							0	Sallevelt et al., 2013
PIVc1e5	0	0													0	Sallevelt et al., 2013
PIVc1e6	0	0	0	0	0										0	Sallevelt et al., 2013
PIVc1e7	0	0													0	Sallevelt et al., 2013
PIVc1e8	0	0													0	Sallevelt et al., 2013
PIVc2e1	100	100	100	100	100	100	100								0	Sallevelt et al., 2013
PIVc2e2	0	0	0												0	Sallevelt et al., 2013

Embryos	Blas. 1	Blas. 2	Blas. 3	Blas. 4	Blas. 5	Blas. 6	Blas. 7	Blas. 8	Blas. 9	Blas. 10	Blas. 11	Blas. 12	Blas. 13	Whole Embryo	SD	References
PIVc2e3	0	0													0	Sallevelt et al., 2013
PIVc2e4	0	0	0	0	0	0	0								0	Sallevelt et al., 2013
PIVc2e5	0	0													0	Sallevelt et al., 2013
PIVc2e6	0	0	0	0	0	0	0	0	0						0	Sallevelt et al., 2013
PIVc2e7	0	0													0	Sallevelt et al., 2013
PIVc2e8	0	0	0												0	Sallevelt et al., 2013
PIVc2e9	0	0	0	0	0	0									0	Sallevelt et al., 2013
PIVc2e11	0	0	0	0	0	0	0								0	Sallevelt et al., 2013
PIVc2e12	0	0	0	0	0										0	Sallevelt et al., 2013
PIVc2e13	0	0	0	0	0										0	Sallevelt et al., 2013
PIVc1e2	31	31	29											27	1.91	Sallevelt et al., 2013
1	0	0													0	Steffann et al., 2006
2	100	100												100	0	Steffann et al., 2006
3	0	0													0	Steffann et al., 2006
MT-ATP6: m.9185T>C (*Gene*: Mutation)																
2	0	0	0	0	0										0	unpublished
3	54	53	54												0.58	unpublished
4	5	4	3												1	unpublished
1	18	20													1.41	unpublished
MT-ND3: m.10197G>A (*Gene*: Mutation)																
1	10	10												10	0	unpublished
2	90	90												90	0	unpublished
3	100	100												100	0	unpublished
MT-TK: m.8344A>G (*Gene*: Mutation)																
2	62	63												63	0.58	unpublished
1	95	95												96	0.58	unpublished
7	54	52													1.41	unpublished
9	60	58													1.41	unpublished
4	95	94												97	1.53	unpublished

Embryos	Blas. 1	Blas. 2	Blas. 3	Blas. 4	Blas. 5	Blas. 6	Blas. 7	Blas. 8	Blas. 9	Blas. 10	Blas. 11	Blas. 12	Blas. 13	Whole Embryo	SD	References
3	70	72												74	2	unpublished
5	57	51												58	3.78	unpublished
8	85													91	4.24	unpublished
6	60	35												51	12.66	unpublished
MT-TL1: m.3243A>G (*Gene*: Mutation)																
1	0	0													0	Monnot et al., 2011
2	0	0													0	Monnot et al., 2011
3	0	0												0	0	Monnot et al., 2011
4	5	5													0	Monnot et al., 2011
7	13	13													0	Monnot et al., 2011
5	7	8													0.71	Monnot et al., 2011
r1	16	17	18	18	19	19									1.17	Monnot et al., 2011
6	12	14													1.41	Monnot et al., 2011
11		25												23	1.41	Monnot et al., 2011
19	63	66												63	1.73	Monnot et al., 2011
r2	22	21	25												2.08	Monnot et al., 2011
14	40	41												37	2.08	Monnot et al., 2011
8	13	16													2.12	Monnot et al., 2011
10	14	15	13	13	14	15	15	16	17	17	17	20	19		2.17	Monnot et al., 2011
12	27	27												23	2.31	Monnot et al., 2011
17	54	55												50	2.65	Monnot et al., 2011
r3	47	48	49	53	53										2.83	Monnot et al., 2011
23	69	71												75	3.06	Monnot et al., 2011
15	51	55												48	3.51	Monnot et al., 2011
22	72	77													3.54	Monnot et al., 2011
9	17	22												15	3.61	Monnot et al., 2011
21	67	68												74	3.79	Monnot et al., 2011
18	55	55												62	4.04	Monnot et al., 2011
20	55	55												62	4.04	Monnot et al., 2011

Embryos	Blas. 1	Blas. 2	Blas. 3	Blas. 4	Blas. 5	Blas. 6	Blas. 7	Blas. 8	Blas. 9	Blas. 10	Blas. 11	Blas. 12	Blas. 13	Whole Embryo	SD	References
16	39	40												50	6.08	Monnot et al., 2011
13	15	21												30	7.55	Monnot et al., 2011
PIc2e4	30	30													0	Sallevelt et al., 2013
PIc2e5	1	1													0	Sallevelt et al., 2013
PIIc1e3	2	2													0	Sallevelt et al., 2013
PIIc2e5	56	57	57												0.58	Sallevelt et al., 2013
PIIIc1e10	41	41												40	0.58	Sallevelt et al., 2013
PIc1e1	4	5													0.71	Sallevelt et al., 2013
PIc3e2	49	50	48	48	48	48	48	48	48						0.71	Sallevelt et al., 2013
PIIIc1e8	35	36													0.71	Sallevelt et al., 2013
PIIIc1e7	60	61	62												1	Sallevelt et al., 2013
PIc2e2	41	43	42	43	44	43	45	43	44						1.17	Sallevelt et al., 2013
PIIIc1e6	70	70	67	68	70	67	68	69							1.30	Sallevelt et al., 2013
PIIIc1e4	62	63	60	60										60	1.41	Sallevelt et al., 2013
PIIc1e2	25	27	25	25	26	22	25	26							1.46	Sallevelt et al., 2013
PIIIc1e3	19	21	22	22	20	22	20	20	21	21	25				1.60	Sallevelt et al., 2013
PIIc1e1	49	54	50	48	51	51	50	50	48	50	50				1.64	Sallevelt et al., 2013
PIIc1e6	10	11	11	11	11	11	11	5	10	12					1.95	Sallevelt et al., 2013
PIc1e2	61	56	58	59	56	56									2.07	Sallevelt et al., 2013
PIIc2e4	29	31	27	32										28	2.07	Sallevelt et al., 2013
PIIIcIe1	51	53												48	2.52	Sallevelt et al., 2013
PIc2e3	40	43	42	48	43	44	43	48	45						2.64	Sallevelt et al., 2013
PIIc2e3	36	38	32	34	33	28	34	33	33	26	34	33			3.19	Sallevelt et al., 2013
PIc3e6	40	38	43	33	41	37	40								3.24	Sallevelt et al., 2013
PIc3e3	44	42	41	42	44	35	45								3.34	Sallevelt et al., 2013
PIc3e1	60	61	62	58	58	59	50								3.95	Sallevelt et al., 2013
PIc3e4	60	50	59	66	62									63	5.48	Sallevelt et al., 2013
PIIc1e4	21	38	20	20	22	26	20	20	20	22					5.63	Sallevelt et al., 2013
PIc3e5	44	44	54	50	44	52	42	35	50						5.92	Sallevelt et al., 2013
PIIc2e1	59	56	56	54	54	57	56	77	56	55	57	56			6.21	Sallevelt et al., 2013

Embryos	Blas. 1	Blas. 2	Blas. 3	Blas. 4	Blas. 5	Blas. 6	Blas. 7	Blas. 8	Blas. 9	Blas. 10	Blas. 11	Blas. 12	Blas. 13	Whole Embryo	SD	References
PIIc1e5	42	54	54												6.93	Sallevelt et al., 2013
PIIc2e2	26	27	30	23	23	5	24	23						23	7.05	Sallevelt et al., 2013
PIc2e1	60	82	62	62	62										9.21	Sallevelt et al., 2013
PIIIc2e1	69	71	70	60	73									70	4.53	Sallevelt et al., 2013
PIIIc2e2	18	14	17											17	1.73	Sallevelt et al., 2013
embryo 3	21	22	24	21	21	21	21	25	23	20	23	18			1.87	Vandewoestyne et al., 2012
embryo 7	52	49	38	54	45	58									7.09	Vandewoestyne et al., 2012
1	7	7.6	5.1	6.3	7.9	8	7.2								1.03	Treff et al., 2012
2	86	85													0.71	Treff et al., 2012
3	68	70	68	70	69	67	67								1.27	Treff et al., 2012
4	47	53	53	53	53	56									2.95	Treff et al., 2012
5	26	22	23	26	30	22									3.13	Treff et al., 2012

Figure S1. Test Blastomere Heteroplasmy and Mutation Load in Preimplantation Embryos

(A and B) The heteroplasmy value in one randomly chosen blastomere (chosen with use of a random number generator) was compared to the heteroplasmy in each remaining cell of the corresponding rhesus macaque embryo (A; data are from 22 embryos in Lee et al., 2012) or human embryo (B; data are from 105 preimplantation embryos in Steffann et al., 2006; Monnot et al., 2011; Treff et al., 2012; Vandewoestyne et al., 2012; Sallevelt et al., 2013; and unpublished data). The line is a linear fit to the data.

(C) Blas., blastomere; SD, standard deviation. The mutant load expresses the ratio of mutant to total (mutant + wild-type) mtDNA. Values were achieved through semiquantitative fluorescent PCR followed by restriction enzyme digestion (Monnot et al., 2011) in a single blastomere (blas.), multiple blastomeres (grey shading), and the remaining embryo (whole embryo) for a given preimplantation embryo. The SD corresponds to the dispersion of the heteroplasmy values among different samples from a given embryo.